

FORM 2
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COMPLETE SPECIFICATION
(See Sections 10 and Rule 13)

1. TITLE OF THE INVENTION

DYNAMIC SAMPLING MOIRÉ DISPLACEMENT SENSOR

2. APPLICANT(S)

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3. PREAMBLE TO THE DESCRIPTION

COMPLETE SPECIFICATION

The following specification particularly describes the invention and the manner in which it is to be performed.

TECHNICAL FIELD

[0001] The present disclosure relates to non-contact displacement sensing systems. More particularly, the present invention relates to a Dynamic Sampling Moiré - based linear displacement measurement system employing an
5 electronically tunable display as grating with a CMOS linear image sensor for achieving both high-resolution and long-range displacement measurement within a single optical configuration.

BACKGROUND OF THE INVENTION

[0002] Linear displacement measurement is widely used in industrial and
10 scientific applications, including robotics, precision positioning systems and semiconductor manufacturing. Accurate displacement sensing is important for position measurement and motion control.

[0003] Various displacement measurement technologies have been developed, including mechanical encoders, capacitive sensors, magnetic encoders, laser
15 interferometers, and optical encoder systems. Among these, optical displacement measurement systems are widely utilized due to their non-contact operation, high sensitivity, and ability to provide accurate position information over a range of operating conditions.

[0004] Many optical displacement measurement systems employ a scale and
20 readhead arrangement in which periodic optical features are detected and processed to determine relative displacement between two components. Such systems commonly utilize fixed optical gratings, encoder scales, or Moir'e fringe generating elements to produce signals representative of displacement.

[0005] Sampling Moir'e-based displacement sensing methods use periodic
25 grating patterns to measure displacement through phase analysis. The captured fringe image is processed to extract phase information corresponding to the relative displacement. The measurement range of such systems is generally determined by the grating pitch and the associated optical arrangement.

[0006] A limitation of conventional grating-based and Sampling Moir'e

displacement sensing systems is that the grating pitch is generally fixed once the sensing system is manufactured or assembled. Fine-pitch gratings may provide enhanced measurement resolution but can restrict the measurement range.

Conversely, coarse-pitch gratings may extend the measurable displacement range
5 while reducing achievable resolution.

[0007] Conventional displacement sensing systems may employ different grating structures, multiple sensing assemblies, or gratings having multiple pitch regions to achieve different measurement characteristics. However, such grating structures are typically fabricated with fixed geometries, resulting in increased
10 manufacturing complexity and limited adaptability once the system is assembled.

[0008] Accordingly, there exists a need for an optical displacement sensing system capable of adapting its measurement range and resolution through electronic control of grating pitch within a common sensing architecture, without requiring multiple physical gratings or mechanical reconfiguration.

15 **OBJECTS OF THE PRESENT DISCLOSURE**

[0009] The following are some of the technical objectives of the present disclosure, which are addressed and satisfied by at least one of the embodiments described herein:

[0010] A general object of the present disclosure is to achieve high -
20 resolution, long - range relative linear displacement measurements of a targeted surface, thereby overcoming the range limitations of high-resolution physical-grating optical encoders

[0011] Another object of the present system is to execute this measurement by integrating Sampling Moiré processing framework with dynamically generated
25 grating patterns on an electronic display, wherein the spatial pitch of the pattern is electronically varied in real-time to obtain high resolution and long range.

[0012] Yet another object of the present system is to provide imaging optics assembly configured to project and image the electronically generated grating pattern directly onto the linear image sensor pixels.

[0013] Yet another object of the present system is to provide a compact, lightweight and non-contact sensor head for real-time displacement tracking

[0014] These and other objects and advantages will become more apparent when reference is made to the following description and accompanying drawings.

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SUMMARY OF THE INVENTION

[0015] This summary is provided to introduce certain concepts relating to the disclosed dynamic Sampling Moir'e displacement sensing system and is not intended to identify all features of the invention.

10 [0016] In an aspect of the present disclosure, there is provided a non-contact optical displacement sensing system comprising an electronic grating configured to generate programmable grating patterns, a sensor readhead comprising an optical projection assembly and a CMOS linear image sensor, and a data acquisition and processing module configured to determine displacement from Sampling Moir'e fringe signals.

15 [0017] In another aspect of the present disclosure, the electronic grating is configured to generate grating patterns having dynamically selectable spatial pitch. The grating pitch may be varied electronically to provide different measurement characteristics without requiring physical grating replacement.

20 [0018] In another aspect of the present disclosure, displacement information obtained from grating patterns having different spatial pitches may be utilized to provide enhanced measurement range and resolution characteristics within a common sensing architecture.

[0019] In another aspect of the present disclosure, the disclosed system enables adaptation of measurement characteristics through electronic control of
25 grating pitch, thereby reducing or eliminating the need for optical realignment or mechanical reconfiguration.

[0020] Various objects, features, aspects, and advantages of the inventive subject matter will become more apparent from the following detailed description of preferred embodiments, along with the accompanying drawing figures in

which like numerals represent like components.

BRIEF DESCRIPTION OF THE ACCOMPANYING DRAWINGS

[0021] The features and advantages of the present disclosure will be better understood by reference to the accompanying drawings, wherein like reference numerals designate like parts throughout the several views. The following descriptions are intended as exemplary illustrations of specific embodiments of the systems, devices, and methods consistent with the subject matter claimed herein, wherein:

[0022] **FIG. 1A** illustrates an exemplary overview of the dynamic sampling Moiré displacement measurement system depicting the sensor readhead (102), the electronic grating (104), and the processing modules (106);

[0023] **FIG. 1B** illustrates an exemplary cross-sectional view of the sensor readhead (102) detailing the internal arrangement of the lens holders (108), the sensor driver PCB (110), and the linear image sensor (112);

[0024] **FIG. 2A** illustrates an exemplary optical configuration employing a convex lens assembly (202) for projecting a programmable grating pattern onto a linear image sensor, in accordance with an embodiment of the present disclosure;

[0025] **FIG. 2B** illustrates a corresponding exploded perspective view of the convex lens optical configuration shown in **FIG. 2A**;

[0026] **FIG. 3A** illustrates an exemplary optical configuration employing a rod lens array (302) for imaging a programmable grating pattern, in accordance with another embodiment of the present disclosure;

[0027] **FIG. 3B** illustrates a corresponding exploded perspective view of the rod lens array optical configuration shown in **FIG. 3A**;

[0028] **FIG. 4A** illustrates an extracted phase versus displacement plot of the conventional sampling Moiré method, demonstrating the displacement range limitation where the wrapped phase difference limits unambiguous measurement strictly to half of the physical grating pitch ($+p/2$ to $-p/2$), corresponding to a phase window of $(+\pi$ to $-\pi)$;

[0029] **FIG. 4B** illustrates an exemplary block diagram of the multi-pitch phase processing sequence, detailing fine image acquisition (406), coarse image acquisition (408), coarse sampling Moiré processing (410), fine sampling Moiré processing (412), coarse-fine pattern combination (414), and final displacement measurement (416); and

[0030] **FIG. 5** illustrates experimental validation results of the proposed dynamic sampling Moiré displacement sensor, including a performance comparison between displacement measurements obtained from the present disclosure and a high-precision reference encoder.

[0031] It should be understood that the drawings are intended to represent exemplary embodiments of the present disclosure for purposes of illustration. Those skilled in the art will recognize that modifications, alternative configurations, optical arrangements, sensing architectures, and displacement tracking methods may be employed herein without departing from the core principles of the disclosure.

DETAILED DESCRIPTION

[0032] The following detailed description is provided with reference to the accompanying drawings to enable a clear understanding of the disclosed embodiments. The described embodiments are illustrative only and are not intended to limit the scope of the present disclosure. Various modifications, equivalents, and alternative implementations may be made without departing from the spirit and scope of the invention as defined by the appended claims.

[0033] Certain specific details are described to provide a thorough understanding of the disclosed embodiments. However, persons skilled in the art will appreciate that the invention may be practiced without all of the specific details described herein. Furthermore, individual features described in connection with one embodiment may be used independently or in combination with features of other embodiments.

[0034] Individual embodiments may be described with reference to flowcharts, flow diagrams, data flow diagrams, structure diagrams, or block diagrams. The

illustrated sequence of operations is provided for explanation and should not be construed as limiting, as operations may be performed in different orders, in parallel, concurrently, or with additional steps not explicitly shown. Processes described herein may correspond to methods, functions, procedures, subroutines, or program modules, and equivalent implementations are considered within the scope of the present disclosure.

[0035] The terms “exemplary” and “demonstrative” are used herein solely to illustrate example embodiments and should not be construed as limiting or as indicating a preferred embodiment. Equivalent structures, arrangements, and techniques known to persons skilled in the art are also contemplated.

Furthermore, the terms “includes,” “including,” “has,” “having,” “contains,” and similar expressions are intended to be inclusive in a manner similar to the term “comprising,” such that additional elements or features may be present without departing from the scope of the present disclosure.

[0036] Reference herein to “one embodiment” or “an embodiment” indicates that a particular feature, structure, or characteristic is included in at least one embodiment of the present disclosure. Such references are not necessarily to the same embodiment, and features described in connection with different embodiments may be combined in any suitable manner.

[0037] The terminology used herein is for descriptive purposes and is not intended to limit the scope of the present disclosure. Unless the context clearly indicates otherwise, singular terms such as “a,” “an,” and “the” include plural references. The terms “comprising,” “including,” and similar expressions are intended to be inclusive and do not exclude the presence of additional features, elements, or steps. Furthermore, the term “and/or” includes any one or more combinations of the associated listed items.

[0038] The present disclosure provides a dynamic sampling Moiré displacement sensing system and method for non-contact linear displacement measurement. Referring to Figures FIG. 1A and FIG. 1B, the system comprises an electronic grating (104) configured to generate programmable linear grating patterns, a sensor readhead (102), and a data acquisition and processing module

(106) configured to determine displacement from acquired image data.

[0039] As illustrated in Figures FIG. 1A and FIG. 1B, the sensor readhead (102) comprises one or more lens holders (108), a sensor driver PCB (110), and a CMOS linear image sensor (112). The electronic grating (104) and the sensor readhead (102) are arranged such that a generated grating pattern is projected onto the active pixel array of the CMOS linear image sensor (112).

[0040] The electronic grating (104) generates programmable periodic linear grating patterns. The spatial pitch of the generated grating pattern may be varied electronically through software control without requiring replacement of a physical grating element.

[0041] The CMOS linear image sensor (112) captures intensity distributions corresponding to the projected grating pattern. Relative displacement between the electronic grating (104) and the sensor readhead (102) causes corresponding phase variations in the captured fringe image. The sensor driver PCB (110) controls image acquisition and transfers image data to the data acquisition and processing module (106).

[0042] Referring to Figures FIG. 2A and FIG. 2B, an embodiment of the sensor readhead utilizes a convex lens assembly (202). The convex lens (202) projects and focuses the grating pattern generated by the electronic grating (104) onto the CMOS linear image sensor (112). The focal length and placement of the convex lens may be selected to achieve a desired imaging magnification, field of view, and measurement range.

[0043] Referring to Figures FIG. 3A and FIG. 3B, an alternative embodiment utilizes a rod lens array (302). The rod lens array (302) provides image projection and focusing functionality while reducing the physical size of the sensor readhead (102). Such a configuration may be advantageous for compact sensing systems and applications having limited installation space.

[0044] Referring to Figure FIG. 4A, a conventional sampling Moiré approach may exhibit limitations arising from the selection of a fixed grating pitch.

Fine-pitch gratings may provide high displacement resolution but can restrict the measurable displacement range, whereas coarse-pitch gratings may increase the

measurable range at the expense of resolution. Consequently, conventional systems often require a compromise between measurement range and measurement resolution.

[0045] Referring to Figure FIG. 4B, the disclosed dynamic Sampling Moiré

5 methodology utilizes multiple electronically generated grating pitches to overcome limitations associated with conventional fixed-pitch Sampling Moiré systems. In various embodiments, selection of grating pitch may be predetermined, user-configurable, automatically selected based on measurement requirements, or dynamically adjusted during operation. The processing module
10 may determine an appropriate grating pitch according to desired measurement range, resolution, signal quality, or operating conditions. A coarse grating pattern may be generated at step (404) and a corresponding fringe image acquired at step (408). Similarly, a fine grating pattern may be generated at step (402) and a corresponding fringe image acquired at step (406).

15 **[0046]** Sampling Moiré signals are generated from the acquired images at steps (410) and (412). The coarse-pitch Sampling Moiré signal generally provides an extended unambiguous displacement range, whereas the fine-pitch Sampling Moiré signal provides enhanced displacement sensitivity and resolution. The displacement information obtained from the coarse and fine Sampling Moiré
20 signals is combined at step (414). The coarse-pitch displacement estimate may be utilized to determine a displacement interval, while the fine-pitch displacement estimate may be utilized to refine the displacement measurement within the determined interval.

[0047] A final displacement estimate is generated at step (416). By combining
25 displacement information obtained from multiple electronically generated grating pitches, the disclosed system enables dynamic adaptation of measurement characteristics while a single sensing platform may support multiple measurement ranges and resolutions.

[0048] The data acquisition and processing module (106) may perform signal
30 conditioning, noise reduction, Fourier-domain phase extraction, phase unwrapping, correlation analysis, interpolation, filtering, calibration correction,

and displacement estimation operations. The processing module may comprise one or more microprocessors, micro-controllers, digital signal processors, field-programmable gate arrays, or any other computing devices.

[0049] The disclosed system enables adaptation of measurement

5 characteristics through electronic selection of grating pitch. Fine-pitch grating patterns may be utilized for enhanced displacement resolution, whereas coarse-pitch grating patterns may be utilized for extended measurement range. Multiple grating pitches may additionally be combined to achieve both enhanced resolution and extended measurement range.

10 **[0050]** The disclosed architecture eliminates the requirement for physical grating replacement, optical realignment, lens repositioning, or mechanical reconfiguration when transitioning between different measurement modes. Accordingly, a single sensing platform may support multiple measurement ranges and resolutions through electronic control of the generated grating pattern.

15 **[0051]** Although the illustrated embodiments describe the use of coarse-pitch and fine-pitch grating patterns, the disclosed methodology is not limited thereto. In various embodiments, two or more grating pitches may be utilized and combined to provide desired measurement characteristics according to a particular application.

20 **[0052]** Referring to Figure FIG. 5, experimental validation was performed to evaluate operation of the disclosed dynamic Sampling Moiré displacement sensing system. Grating patterns having different spatial pitches were generated and corresponding fringe images were acquired and processed to obtain displacement estimates. The obtained results verify generation of Sampling
25 Moiré fringe signals using electronically selectable grating pitches and demonstrate displacement measurement based on phase information extracted from the acquired images.

[0053] In various embodiments, the disclosed system may be employed in robotics, semiconductor manufacturing, vibration analysis, structural monitoring,
30 inspection systems, scientific instrumentation, positioning systems, and other applications requiring non-contact displacement measurement.

Exemplary Embodiments of the Invention

[0054] In an exemplary embodiment, the electronic grating (104) comprises a programmable display device configured to generate periodic linear grating patterns having electronically selectable spatial pitch. The programmable display
5 device may comprise an LCD display, OLED display, microLED display, digital micromirror device (DMD), spatial light modulator (SLM), electronic paper display, or another optical modulation device capable of generating periodic optical structures.

[0055] In another exemplary embodiment, the CMOS linear image sensor
10 (112) comprises a linear image sensor configured for high-speed acquisition of fringe intensity distributions corresponding to the generated grating patterns.

[0056] In another exemplary embodiment, the optical projection assembly comprises a convex lens (202), a rod lens array (302), or combinations thereof for projecting the generated grating pattern onto the active pixel array of the CMOS
15 linear image sensor (112).

[0057] In another exemplary embodiment, grating pitch is selected automatically by the data acquisition and processing module (106) according to a desired measurement range, resolution requirement and performs one or more signal processing operations including thinning out, interpolation, phase
20 extraction, phase unwrapping, calibration correction, and displacement estimation.

[0058] In another exemplary embodiment, the grating pitch generated by the electronic grating (104) is dynamically varied under software control to provide different measurement characteristics. Fine-pitch gratings may be utilized to
25 improve displacement resolution, while coarse-pitch gratings may be utilized to increase measurement range.

[0059] In another exemplary embodiment, displacement information obtained from two or more grating pitches is combined to reduce measurement ambiguity and provide improved displacement estimation across a wider operating range.

30 [0060] In another exemplary embodiment, the disclosed system operates

without physical grating replacement, optical realignment, lens repositioning, or mechanical reconfiguration during transition between measurement modes.

Advantages -

- 5 • The present invention enables dynamic electronic variation of grating pitch, thereby eliminating the need for fabrication, replacement, or alignment of multiple pitch physical grating structures.
- 10 • The present invention enables a single sensing platform to support both high-resolution displacement measurement and extended measurement range through software-controlled selection and combination of grating pitches.
- The present invention reduces measurement ambiguity by combining displacement information obtained from multiple Sampling Moiré signals generated using different grating pitches.

15 **[0061]** The disclosed dynamic Sampling Moiré displacement sensing system may be implemented using different display technologies, optical assemblies, image sensing devices, processing architectures, calibration methodologies, and fringe analysis techniques. Such variations are considered to fall within the scope of the present disclosure provided that they employ the principles described herein.

20 **[0062]** Numerous modifications and adaptations of the disclosed invention will be apparent to those skilled in the art. It is therefore intended that the appended claims encompass all such modifications, variations, and equivalents as fall within the true spirit and scope of the present invention.

We Claim

1. A system for non-contact linear displacement measurement, comprising:

- an electronic grating (104) configured to generate programmable grating patterns having selectable spatial pitch;
- 5 • a sensor readhead (102) comprising a lens assembly configured to project the generated grating pattern onto a CMOS linear image sensor (112); and
- 10 • a data acquisition and processing module (106) configured to acquire image data from the CMOS linear image sensor (112) and determine relative linear displacement based on phase variations of generated Sampling Moiré fringe signals.

2. The system as claimed in claim 1, wherein the data acquisition and processing module (106) determines displacement using a Sampling Moiré Method.

15 3. The system as claimed in claim 1, wherein the electronic grating (104) is configured to dynamically vary the spatial pitch of the generated grating pattern through electronic or software control.

4. The system as claimed in claim 3, wherein the electronic grating (104) is configured to generate a plurality of grating patterns having different spatial
20 pitches for obtaining displacement measurements having different measurement characteristics.

5. The system as claimed in claim 4, wherein displacement information obtained from at least two different grating pitches is combined to improve displacement estimation, extend measurement range, reduce ambiguity, or
25 enhance measurement resolution.

6. The system as claimed in claim 1, wherein the electronic grating (104) comprises at least one of an LCD display, OLED display, microLED display, digital micromirror device (DMD), spatial light modulator (SLM),

electronic paper display, or another programmable optical modulation device.

7. The system as claimed in claim 1, wherein the lens assembly comprises a convex lens (202), a rod lens array (302), or a combination thereof.

5 8. The system as claimed in claim 1, wherein the data acquisition and processing module (106) performs one or more of phase extraction, phase unwrapping, Fourier-domain analysis, correlation analysis, interpolation, filtering, calibration correction, and displacement estimation.

10 9. The system as claimed in claim 1, wherein the system is configured to transition between different measurement ranges and resolution characteristics without physical grating replacement, optical realignment, lens repositioning, or mechanical reconfiguration.

15 10. The system as claimed in claim 1, wherein the sensor readhead (102) further comprises one or more lens holders (108) and a sensor driver PCB (110) configured to control image acquisition from the CMOS linear image sensor (112).

ABSTRACT

DYNAMIC SAMPLING MOIRÉ DISPLACEMENT SENSOR

The present disclosure relates to a system (FIG. 1A) for high-resolution, long-range non-contact linear displacement measurement. The system includes an electronic grating (104) implemented as an electronically tunable display to generate programmable grating patterns with selectable spatial pitch.

Embodiments of the present system covers two imaging optics assembly one with a convex lens (202) another with rod lens array (302), which projects the grating pattern onto a CMOS linear image sensor (112). A processing module (106) in a sensor readhead (102) captures the fringe image to determine displacement via a sampling Moiré framework. By dynamically varying the pitch to alternate between a fine electronic grating pattern (402) and a coarse electronic grating pattern (404), the module executes fine image acquisition (406) and coarse image acquisition (408). The processing module (106) combines them via a coarse-fine pattern combination (414) to yield a final displacement measurement (416). This achieves sub-micron resolution and an extended macro-scale tracking range overcoming the phase wrapping limitations ($+p/2$ to $-p/2$) of conventional fixed-pitch sampling Moiré sensors.

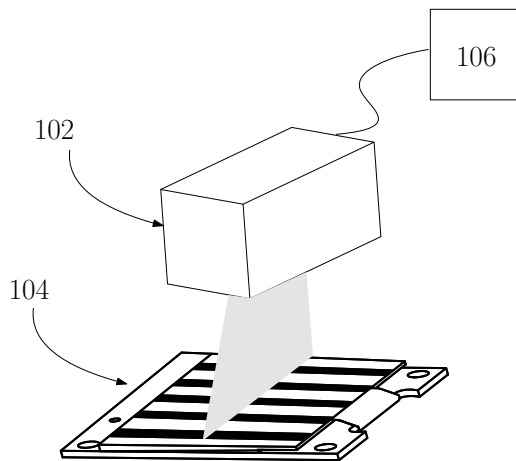


FIG. 1A

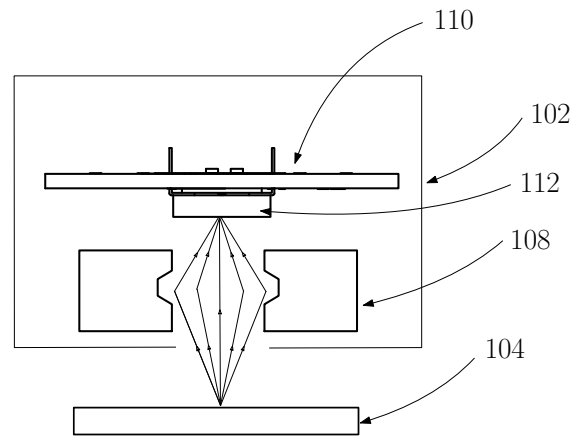


FIG. 1B